

Maintenance Schedule

Diesel Engine

EPA/CARB

10V1600C60

12V1600C60

Application Group 5B

MSE50169/01E



Power. Passion. Partnership.

Engine model	kW/cyl.	Application group
10V1600C60	57 kW/cyl.	5B, Continuous operation, variable
12V1600C60	57 kW/cyl.	5B, Continuous operation, variable

Table 1: Applicability

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1 Maintenance Schedule

1.1 Preface

The maintenance schedule is applicable to products certified in accordance with U.S. emissions regulations and operated in the area of applicability of these regulations.

Information about the latest maintenance schedule is available at: <http://www.mtu-online.com>.

The following instructions must be observed for these products:

The maintenance schedule is an integral part of the emissions certification. Nonobservance of the maintenance instructions can result in violation of the statutory emission specifications. Modification or removal of mechanical or electronic components or the installation of additional components as well as the execution of calibration processes that might affect the emission characteristics of the product are prohibited by emission regulations. Emission-related components, e.g. control units, sensors, cylinder heads, injectors and exhaust flaps can only be serviced, replaced or repaired if components approved by MTU or equivalent components are used.

The maintenance system for MTU products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals and the scope of required maintenance activities are the result of operational experience and therefore represent recommendations. Severe operating and ambient conditions may require additional maintenance work and/or modification of the maintenance intervals.

Maintenance activities are specified with intervals based on hours in operation and time limits. Whichever value is reached first shall apply.

The individual maintenance intervals are classified in accordance with the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product.

QL2: Exchange of components and parts (corrective only, not part of the maintenance schedule).

QL3: Maintenance work which requires partial disassembly of the product.

QL4: Maintenance work which requires complete disassembly of the product.

Additional notes on maintenance and preservation:

For change intervals for fluids and lubricants as well as for preservation specifications, refer to the relevant instructions of the component manufacturers and the MTU Fluids and Lubricants Specifications. The latest version for drive systems is available online at: <http://www.mtu-online.com>, and for power generation systems at: <http://www.mtuonsiteenergy.com>.

Components which are not listed in this maintenance schedule must be serviced in accordance with Manufacturer's Instructions.

1.2 Allocation matrix application group 5B

Load profile		10/12V1600C60/A60						
Load %	Time %	Load factor %	Load indicator %	0 - 60 kW/cyl.				
110 ₍₁₀₀₋₁₁₀₎	—	0 - 31	≤ 17	21,000	—	—	—	Standard
100 ₍₉₀₋₁₀₀₎	12							
90 ₍₈₀₋₉₀₎	12							
80 ₍₆₅₋₈₀₎	—							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	16							
Idle ₍₀₋₅₎	60							
110 ₍₁₀₀₋₁₁₀₎	—	32 - 35	≤ 19	18,000	—	—	—	
100 ₍₉₀₋₁₀₀₎	17							
90 ₍₈₀₋₉₀₎	—							
80 ₍₆₅₋₈₀₎	13							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	15							
Idle ₍₀₋₅₎	55							
110 ₍₁₀₀₋₁₁₀₎	—	36 - 45	≤ 32	15,000	—	—	—	
100 ₍₉₀₋₁₀₀₎	26							
90 ₍₈₀₋₉₀₎	15							
80 ₍₆₅₋₈₀₎	—							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	12							
Idle ₍₀₋₅₎	47							
110 ₍₁₀₀₋₁₁₀₎	—	46 - 55	≤ 41	15,000	—	—	—	
100 ₍₉₀₋₁₀₀₎	35							
90 ₍₈₀₋₉₀₎	15							
80 ₍₆₅₋₈₀₎	—							
65 ₍₅₀₋₆₅₎	—							
50 ₍₃₀₋₅₀₎	—							
30 ₍₅₋₃₀₎	15							
Idle ₍₀₋₅₎	35							

TM-ID: 000005 583 - 003

Table 2: Allocation matrix application group 5B

10/12V1600C60/A60							
Load profile		Load factor %	Load indicator %	0 - 60 kW/cyl.			
Load %	Time %						
110 ₍₁₀₀₋₁₁₀₎	—	56 - 65	≤ 51	12,000	—	—	—
100 ₍₉₀₋₁₀₀₎	44						
90 ₍₈₀₋₉₀₎	17						
80 ₍₆₅₋₈₀₎	—						
65 ₍₅₀₋₆₅₎	—						
50 ₍₃₀₋₅₀₎	—						
30 ₍₅₋₃₀₎	15						
Idle ₍₀₋₅₎	24						
110 ₍₁₀₀₋₁₁₀₎	—	66 - 75	≤ 61	9,000	—	—	—
100 ₍₉₀₋₁₀₀₎	54						
90 ₍₈₀₋₉₀₎	17						
80 ₍₆₅₋₈₀₎	—						
65 ₍₅₀₋₆₅₎	—						
50 ₍₃₀₋₅₀₎	—						
30 ₍₅₋₃₀₎	15						
Idle ₍₀₋₅₎	14						

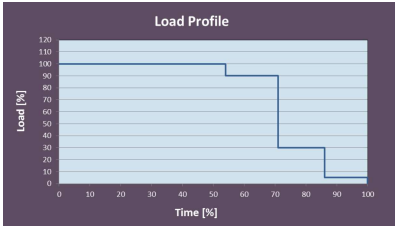
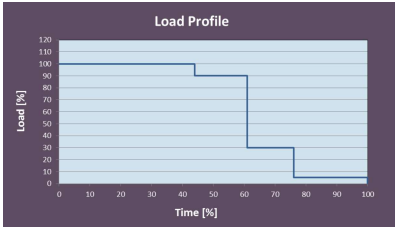


Table 3: Allocation matrix application group 5B

1.3 Maintenance schedule matrix 21,000 h

0-21,000 Operating hours

Task	Limit	Operating hours [h]																													
		Daily	1,000	2,000	3,000	4,000	5,000	5,250	6,000	7,000	8,000	9,000	10,000	10,500	11,000	12,000	13,000	14,000	15,000	15,750	16,000	17,000	18,000	19,000	20,000	21,000					
Engine																															
W0800	-																														
W1008	2 a																														
W0500	-	X																													
W0501	-	X																													
W0506	-	X																													
W0507	-	X																													
W1001	2 a		X	X	X	X	X		X	X	X	X	X		X	X	X	X	X		X	X	X	X	X	X	X				
W1675	2 a		X	X	X	X	X		X	X	X	X	X		X	X	X	X	X		X	X	X	X	X	X	X				
W1019	-			X		X			X		X		X			X		X			X		X		X						
W1716	-				X				X		X				X		X				X		X		X						
W1207	2 a		X		X				X		X				X			X				X			X						
W1003	2 a				X				X		X				X		X				X		X		X						
W8638	-							X					X						X						X						
W1525	6 a							X					X					X							X						
W1526	6 a							X					X					X							X						
W1006	9 a									X								X							X						
W1636	9 a								X									X							X						
W1298	9 a								X									X							X						
W1178	9 a								X									X							X						
W1660	-												X												X						
W1812	5 a												X												X						
W8637	6 a												X												X						
W1055	6 a												X												X						
W1041	9 a												X												X						
W1672	9 a																								X						
W3148	18 a																								X						
w = weeks m = months a = years																															

1.4 Maintenance tasks 21,000 h

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
Engine						
QL 1	-	-	EPA label	Check if emission label is fitted and legible, and if it contains the correct information.		W0800
QL 1	-	2 a	Engine oil filter	Fit new engine oil filters each time the engine oil is changed or, at the latest, on expiry of the time limit (given in years).		W1008
QL 1	Daily	-	Engine operation	Check engine oil level.		W0500
				Carry out visual inspection of engine for general condition and leaks.		W0501
				Check for abnormal running noises, exhaust gas color and vibration.		W0506
				Drain off water and contamination from fuel prefilter.	X	W0507
QL 1	1000	2 a	Fuel filter	Fit new fuel filter or new fuel filter insert.		W1001
QL 1	1000	2 a	Fuel prefilter	Fit new fuel prefilter or new fuel prefilter insert.	X	W1675
QL 1	2000	-	Coupling	Check radial clearance.	X	W1019
QL 1	3000	-	Battery charging alternator	Clean battery-charging generator.		W1716
QL 1	3000	2 a	Valve gear	Check valve clearance, adjust if required. ATTENTION! First adjustment after 1,000 hours.		W1207
QL 1	3000	2 a	Belt drive	Check belt condition and tension Fit new belt(s) if necessary.		W1003
QL 1	5250	-	EGR cooler	Check EGR cooler, replace if necessary.		W8638
QL 1	5250	6 a	NOx sensor	Fit new sensor.		W1525
QL 1	5250	6 a	Oxygen sensor	Fit new sensor.		W1526
QL 3	7000	9 a	Fuel injectors	Replace fuel injectors.		W1006
QL 3	7000	9 a	Engine governor	Reset parameters of drift correction (CDC) and enter injector coding (IIG).		W1636
QL 3	7000	9 a	HP fuel lines	Fit new HP fuel lines.		W1298
QL 3	7000	9 a	Pressure pipe neck (bayonet)	Fit new pressure pipe neck (bayonet) in cylinder head.		W1178
QL 3	10500	-	Air ducting	Clean air ducting.		W1660
QL 3	10500	5 a	Engine mounts	Check for cracks. Measure height of rubber element.		W1812
QL 3	10500	6 a	Rubber sleeves	Check all rubber sleeves, replace if necessary.		W8637
QL 3	10500	6 a	Crankcase breathers	Fit new oil separators.		W1055
QL 3	10500	9 a	Turbochargers	Fit new turbochargers.		W1041
QL 3	21000	9 a	Engine mounts	Fit new rubber elements of engine mounts.		W1672
QL 4	21000	18 a	Extended component maintenance	Overhaul or replace engine.		W3148

w = weeks
m = months
a = years